Q1 (2) - Multiplication and Transpose of Matrix

$A = \operatorname{diag}\{2, -1, 3\}$

and $B = \operatorname{diag}\{-1, 3, 2\}$

$\Rightarrow A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$\Rightarrow B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Then, $A^2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

$$A^2B = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$
$$= \operatorname{diag}\{-4, 3, 18\}$$

Q2 (2) - Multiplication and Transpose of Matrix

$\because B^T A^T = (AB)^T$

$= \left(\begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix} \right)^T$

$= \begin{bmatrix} 8-7 & 2-2 \\ -28+28 & -7+8 \end{bmatrix}^T$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

Hence, $B^T A^T$ is an identity matrix.

Q3 (1) - Multiplication and Transpose of Matrix

If $AB = 0$

$$\Rightarrow \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} \cos^2\theta\cos^2\phi + \sin\theta\sin\phi\cos\theta\cos\phi & \cos^2\theta\cos\phi\sin\phi + \cos\theta\sin\theta\sin^2\phi \\ \cos\theta\cos^2\phi + \sin^2\theta\cos\phi\sin\phi & \cos\theta\sin\theta\cos\phi\sin\phi + \sin^2\theta\sin^2\phi \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence, $\theta - \phi$ is an odd multiple of $\frac{\pi}{2}$

Q4 (1) - Multiplication and Transpose of Matrix

$\because AB = A$ and $BA = B$

$\therefore \quad B^2 = (BA)(BA)$

$= B(AB)A = B \cdot A \cdot A$

$= (BA) \cdot A = BA = B$

Q5 (1) - Multiplication and Transpose of Matrix

$\because A^T = A$ and $B^T = B$

Now, $\quad (ABA)^T$

$A^T(AB)^T = A^T\left(B^T A^T\right) = A(BA) = ABA$

Q6 (1) - Multiplication and Transpose of Matrix

$Q^2 = Q \cdot Q = PAP^T \cdot PAP^T$

$= PA\left(P^T \cdot P\right)AP^T = PA \cdot I \cdot AP^T = PA^2P^T$

Similarly, $Q^3 = PA^3P^T$

$Q^{2005} = PA^{2005}P^T$

Now, $X = P^TQ^{2005}P$

$= P^T\left(PA^{2005}P^T\right)P$

$= \left(P^TP\right)A^{2005}\left(P^TP\right)$

$= 1 \cdot A^{2005} \cdot I = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

Q7 (2) - Multiplication and Transpose of Matrix

$$\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & -\cos\alpha\sin\alpha + \sin\alpha\cos\alpha \\ -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha & \sin^2\alpha + \cos^2\alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

$\because$ If A is an orthogonal matrix, then $AA^T = 1$.

$\therefore \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ is an orthogonal matrix.

Q8 (2) - Properties of Determinants

$$\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$$

On applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$= \begin{vmatrix} a-b+b-c+c-a & b-c & c-a \\ x-y+y-z+z-x & y-z & z-x \\ p-q+q-r+r-p & q-r & r-p \end{vmatrix} = \begin{vmatrix} 0 & b-c & c-a \\ 0 & y-z & z-x \\ 0 & q-r & r-p \end{vmatrix}$$

$= 0$

Q9 (4) - Properties of Determinants

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

$\Rightarrow \quad (x+1)\left(x^2+6x+8-15\right) + 3(10-2x-8) + 5(6-2x-4) = 0$

$\Rightarrow \quad (x+1)\left(x^2+6x-7\right) + 3(2-2x) + 5(2-2x) = 0$

$\Rightarrow \quad x^3 + 6x^2 - 7x + x^2 + 6x - 7 + 6 - 6x + 10 - 10x = 0$

$\Rightarrow \quad x^3 + 7x^2 - 17x + 9 = 0$

$\Rightarrow (x-1)^2(x+9) = 0$

$\therefore x = 1, 1, -9$

Q10 (4) - Properties of Determinants

$$\begin{vmatrix} a+b & a+2b & a+3b \\ a+2b & a+3b & a+4b \\ a+4b & a+5b & a+6b \end{vmatrix}$$

On applying $C_3 \rightarrow C_3 - C_2$

$$\begin{vmatrix} a+b & a+2b & b \\ a+2b & a+3b & b \\ a+4b & a+5b & b \end{vmatrix}$$

On applying $C_2 \rightarrow C_2 - C_1$

$$= \begin{vmatrix} a+b & b & b \\ a+2b & b & b \\ a+4b & b & b \end{vmatrix}$$

$= 0$ [$\because C_2$ and C_3 are same]

Q11 (2) - Properties of Determinants

$$f(1) = \begin{vmatrix} -2 & -16 & -78 \\ -4 & -48 & -496 \\ 1 & 2 & 3 \end{vmatrix}$$

$$f(3) = \begin{vmatrix} 0 & 0 & 0 \\ -2 & -32 & -392 \\ 1 & 2 & 3 \end{vmatrix} = 0$$

$$\begin{aligned} \therefore f(1)f(3) + f(3)f(5)f(5)f(1) &= f(1)f(3)\left[1 + \{f(5)\}^2\right] \\ &= 0 \times \left[1 + \{f(5)\}^2\right] \\ &= 0 \end{aligned}$$

Q12 (2) - Properties of Determinants

$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$$

On applying $R_3 \to R_3 - \alpha R_1$

$$\Rightarrow \begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ b & c & -a\alpha^2 - b\alpha \end{vmatrix} = 0$$

On applying $R_3 \to R_3 - R_2$

$$\Rightarrow \begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ 0 & 0 & -a\alpha^2 - 2b\alpha - c \end{vmatrix} = 0$$

On applying $\mathrm{C}_3 \to \mathrm{C}_3 - (\alpha \mathrm{C}_1 + \mathrm{C}_2)$

$$\Rightarrow \begin{vmatrix} a & b & 0 \\ b & c & 0 \\ 0 & 0 & -a\alpha^2 - 2b\alpha - c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & -a\alpha^2 - 2b\alpha - c \\ b & c & 0 \\ a & b & 0 \end{vmatrix} = 0$$

$$\Rightarrow \quad \left(a\alpha^2 + 2b\alpha + c\right)\left(b^2 - ac\right) = 0$$

$\Rightarrow \quad b^2 - ac = 0$

$\therefore \quad a, b, c$ are in GP.

Q13 (3) - Properties of Determinants

$$\begin{vmatrix} 1+\sin^2\theta & \sin^2\theta & \sin^2\theta \\ \cos^2\theta & 1+\cos^2\theta & \cos^2\theta \\ 4\sin 4\theta & 4\sin 4\theta & 1+4\sin 4\theta \end{vmatrix} = 0$$

On applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \begin{vmatrix} 2+4\sin 4\theta & 2+-4\sin 4\theta & 2+4\sin 4\theta \\ \cos^2\theta & 1+\cos^2\theta & \cos^2\theta \\ 4\sin 4\theta & 4\sin 4\theta & 1+4\sin 4\theta \end{vmatrix} = 0 \left[\because \sin^2\theta + \cos^2\theta = 1\right]$$

$$\Rightarrow (2+4\sin\theta) \begin{vmatrix} 1 & 1 & 1 \\ \cos^2\theta & 1+\cos^2\theta & \cos^2\theta \\ 4\sin 4\theta & 4\sin 4\theta & 1+4\sin 4\theta \end{vmatrix} = 0$$

On applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow (2+4\sin\theta) \begin{vmatrix} 1 & 0 & 0 \\ \cos^2\theta & 1 & 0 \\ 4\sin 4\theta & 0 & 1 \end{vmatrix} = 0$$

$\Rightarrow 2(1+2\sin 4\theta) = 0 \Rightarrow \sin 4\theta = \frac{-1}{2}$

Q14 (3) - Properties of Determinants

$\because a = \text{pth term} = m + (p-1)d$

$b = q \text{ th term} = m + (q-1)d$

$c = n \text{ th term} = m + (r-1)d$

where m is the first term and d is the common difference of an AP.

$$\text{So, } \begin{vmatrix} a & p & 1 \\ b & q & 1 \\ c & r & 1 \end{vmatrix} = \begin{vmatrix} m+(p-1)d & p & 1 \\ m+(q-1)d & q & 1 \\ m+(r-1)d & r & 1 \end{vmatrix}$$

On applying $C_1 \rightarrow C_1 + dC_2 - mC_3$

$$= \begin{vmatrix} -d & p & 1 \\ -d & q & 1 \\ -d & r & 1 \end{vmatrix} = -d \begin{vmatrix} 1 & p & 1 \\ 1 & q & 1 \\ 1 & r & 1 \end{vmatrix} = 0 [\because C_1 \text{ and } C_3 \text{ are same}]$$

Q15 (1) - Properties of Determinants

$$U_n=\begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N \end{vmatrix}$$

$$\sum_{n=1}^{N} U_n=\begin{vmatrix} \frac{N(N+1)}{2} & 1 & 5 \\ \frac{N(N+1)(2N+1)}{6} & 2N+1 & 2N+1 \\ \frac{N^2(N+1)^2}{4} & 3N^2 & 3N \end{vmatrix}$$

$$=\frac{N(N+1)}{2}\begin{vmatrix} 1 & 1 & 5 \\ \frac{2N+1}{3} & 2N+1 & 2N+1 \\ \frac{N(N+1)}{2} & 3N^2 & 3N \end{vmatrix}$$

$$=\frac{N(N+1)(2N+1)}{6}\begin{vmatrix} 1 & 1 & 5 \\ \frac{N(N+1)}{6} & 3 & 3 \\ \frac{N^2}{6} & & \end{vmatrix}$$

$$=\frac{N^2(N+1)(2N+1)}{6}\begin{vmatrix} 1 & 1 & 5 \\ 1 & 3 & 3 \\ N+1 & 6N & 6 \end{vmatrix}$$

$$=\frac{N^2(N+1)(2N+1)}{6}[(18-18N)+(3N+3-6)+5(6N-3N-3)]$$

$$=\frac{N^2(N+1)(2N+1)}{6}[18-18N+3N-3+15N-15]$$

$$=\frac{N^2(N+1)(2N+1)}{6}\times 0=0$$

Q16 (3) - Properties of Determinants

$$D_r=\begin{vmatrix} 2^{r-1} & 2\cdot 3^{r-1} & 4\cdot 5^{r-1} \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$$

$$\sum_{r=1}^{n} D_r=\begin{vmatrix} \frac{2^{n-1}}{2-1} & \frac{2(3^n-1)}{3-1} & \frac{4(5^n-1)}{5-1} \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$$

$$=\begin{vmatrix} 2^n-1 & 3^n-1 & 5^n-1 \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$$

$=0 \quad [\because R_1$ and R_3 are same$]$

Q17 (2) - Adjoint and Inverse

$$(AB)^{-1}=B^{-1}A^{-1}$$

Q18 (4) - Adjoint and Inverse

$$A=\begin{bmatrix}1 & 0 & -k\\ 2 & 1 & 3\\ k & 0 & 1\end{bmatrix}$$

A is invertible, when $|A| \neq 0$

$\Rightarrow 1(1)-k(-k) \neq 0$

$\Rightarrow 1+k^2 \neq 0 \Rightarrow k^2 \neq -1$

i.e. A is invertible for all real values of k.

Q19 (3) - Adjoint and Inverse

$\because \quad (A+B)(A-B)=A^2-B^2$
$\Rightarrow A^2-AB+BA-B^2=A^2-B^2$
$\Rightarrow -AB+BA=0$
$\Rightarrow \quad AB=BA$
Now, $\left(ABA^{-1}\right)^2=\left(BAA^{-1}\right)^2=(B \cdot I)^2=(B)^2$

Q20 (4) - Adjoint and Inverse

Adjoint of a symmetric matrix is symmetric matrix. Adjoint of a unit matrix is a unit matrix.

$A(\operatorname{adj} A)=(\operatorname{adj} A)A=|A|I$

Adjoint of a diagonal matrix is a diagonal matrix.

Q21 (1) - Adjoint and Inverse

Suppose $A=\begin{bmatrix}0 & -3 & 1\\ 3 & 0 & 0\\ -1 & 0 & 0\end{bmatrix}$ which is a skew-symmetric matrix.

Then, $|A|=(0)-3(-2)+(-6)$
$=0+6-6=0$

Hence, A is a singular matrix.

Q22 (3) - Adjoint and Inverse

$A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}$

$A^{-1}\frac{1}{|A|}\operatorname{adj}(A)$

$=\frac{1}{6}\begin{bmatrix}6 & -(0) & 0\\ -(0) & 4 & -1\\ 0 & -(-2) & 1\end{bmatrix}$

$\Rightarrow 6A^{-1}=\begin{bmatrix}6 & 0 & 0\\ 0 & 4 & -1\\ 0 & 2 & 1\end{bmatrix}$ ...(i)

Now $A^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}$

$=\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 5\\ 0 & -10 & 14\end{bmatrix}$ ...(ii)

$A^{-1}=\frac{1}{6}\left(A^2+cA+dl\right)$ [given]

$6A^{-1}=A^2+CA+dl$

From Eqs. (i) and (ii), we get

$\Rightarrow\begin{bmatrix}6 & 0 & 0\\ 0 & 4 & -1\\ 0 & 2 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 5\\ 0 & -10 & 14\end{bmatrix}+\begin{bmatrix}c & 0 & 0\\ 0 & c & c\\ 0 & -2c & 4c\end{bmatrix}+\begin{bmatrix}d & 0 & 0\\ 0 & d & 0\\ 0 & 0 & d\end{bmatrix}$

$\Rightarrow\begin{bmatrix}6 & 0 & 0\\ 0 & 4 & -1\\ 0 & 2 & 1\end{bmatrix}=\begin{bmatrix}1+c+d & 0 & 0\\ 0 & -1+c+d & 5+c\\ 0 & -10-2c & 14+4c+d\end{bmatrix}$

$\Rightarrow c+d=5$

and $5+c=-1$

$\Rightarrow c=-6$

and $d=11$

Q23 (1) - Adjoint and Inverse

$\because A^2-A+1=0$

$A^{-1}\cdot A\cdot A-A^{-1}\cdot A+A^{-1}\cdot I=0$

$\Rightarrow I\cdot A-1+A^{-1}=0$

$\therefore A^{-1}=1-A$

Q24 (4) - System of Equations

We have, $2x - y - z = 12$

$x - 2y + z = 4$

$x + y + \lambda z = 4$

Now, $\quad [A : B] = \begin{vmatrix} 2 & -1 & -1 & : & 12 \\ 1 & -2 & 1 & : & 4 \\ 1 & 1 & \lambda & : & 4 \end{vmatrix}$

On applying $R_3 \to 2R_3 - R_1$ and $R_2 \to 2R_2 - R_1$

$= \begin{vmatrix} 2 & -1 & -1 & : & 12 \\ 0 & -3 & 3 & : & -4 \\ 0 & 3 & 2\lambda + 1 & : & -4 \end{vmatrix}$

On applying $R_3 \to R_3 + R_2$

$= \begin{vmatrix} 2 & -1 & -1 & : & 12 \\ 0 & -3 & 3 & : & -4 \\ 0 & 0 & 2\lambda + 4 & : & -8 \end{vmatrix}$

For $\lambda = -2$,

Rank $A \neq \text{Rank}(AB)$

i.e. for $\lambda = -2$, the given system of equations has no solution.

Q25 (3) - System of Equations

$M^2 N^2 \left(M^T N\right)^{-1} \left(M N^{-1}\right)^T$

$= M^2 N^2 N^{-1} \left(M^T\right)^{-1} \cdot \left(N^{-1}\right)^T M^T$

$= M^2 \cdot N \cdot (-M)^{-1} \cdot (-N)^{-1} (-M) = -M^2 \cdot N \cdot (NM)^{-1} \cdot M$

$= -M^2 \cdot N \cdot (MN)^{-1} \cdot M = -M^2 \cdot N \cdot N^{-1} \cdot M^{-1} \cdot M = -M^2$

Q26 (4) - System of Equations

$$Q=\begin{bmatrix}2^2a_{11} & 2^3a_{12} & 2^4a_{13}\\ 2^3a_{21} & 2^4a_{22} & 2^5a_{23}\\ 2^4a_{31} & 2^5a_{32} & 2^6a_{33}\end{bmatrix}$$

$$|Q|=2^2\cdot2^3\cdot2^4\begin{vmatrix}a_{11} & 2a_{12} & 2^2a_{13}\\ a_{21} & 2a_{22} & 2^2a_{23}\\ a_{31} & 2a_{32} & 2^2a_{33}\end{vmatrix}$$

$$=2^9\cdot2\cdot2^2\begin{vmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{vmatrix}$$

$$=2^{12}\cdot2=2^{13}$$

Q27 (4) - System of Equations

Given, $P^T=2P+1\Rightarrow P=2P^T+1$

$\Rightarrow P=2(2P+1)+1$

$\Rightarrow P=4P+3I$

$\Rightarrow \quad 3(P+\Lambda)=0$

$\Rightarrow \quad P+1=0$

$\Rightarrow \quad PX+X=0$

$\therefore \quad PX=-X$

Q28 (4) - Rank of Matrix

Given, $A=\begin{bmatrix}y+a & b & c\\ a & y+b & c\\ a & b & y+c\end{bmatrix}$

$$|A|=\begin{vmatrix}y+a & b & c\\ a & y+b & c\\ a & b & y+c\end{vmatrix}$$

On applying $R_1\rightarrow R_1+R_2+R_3$

$$=\begin{vmatrix}y+a+b+c & b & c\\ y+a+b+c & y+b & c\\ y+a+b+c & b & y+c\end{vmatrix}$$

$$=(y+a+b+c)\begin{vmatrix}1 & b & c\\ 1 & y+b & c\\ 1 & b & y+c\end{vmatrix}$$

$\therefore$ On applying $R_2\rightarrow R_2-R_1$ and $R_3\rightarrow R_3-R_1$

$$= (y+a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & y & 0 \\ 0 & 0 & y \end{vmatrix}$$

A is of rank 3, if $y \neq -(a+b+c)$ and $y \neq 0$

Q29 (3) - Rank of Matrix

Order of A is $m \times 1$ and order of B is $1 \times n$.

Order of AB is $m \times n$

Since, A and B are non-zero matrices.
$\therefore AB$ will be a non-zero matrix.

The matrix AB will have atleast one non-zero element obtained by multiplying corresponding non-zero elements of A and B. All the two -rowed minors of AB clearly vanish.
$\therefore$ Rank of $AB = 1$